

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course:B.Tech

Branch : Common To All Branches

Semester : II

Subject Code & Name: Engineering Mechanics & BTES203

Max Marks: 60

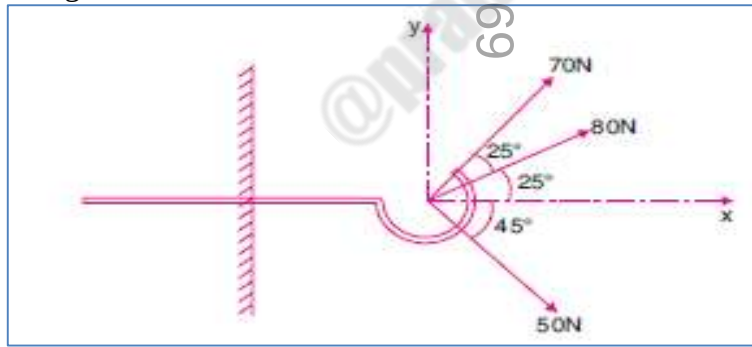
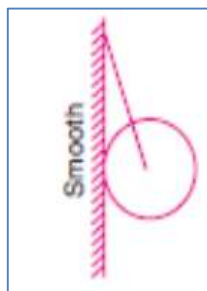
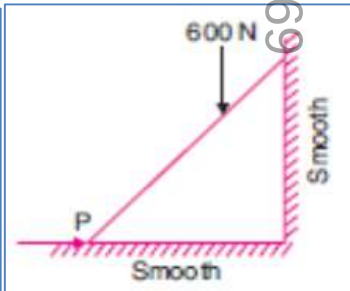
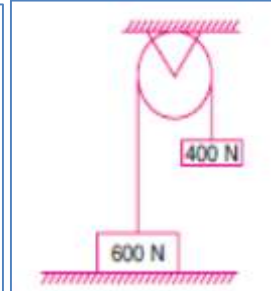
Date:23/12/2024

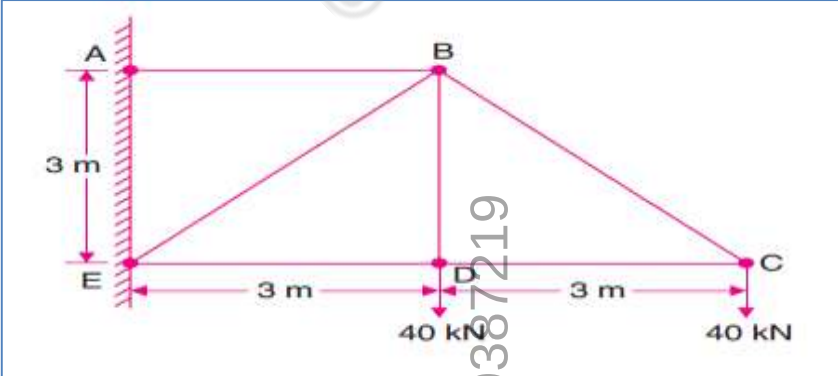
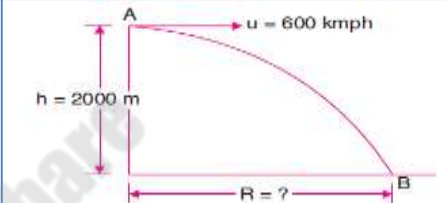
Duration: 3 Hr.

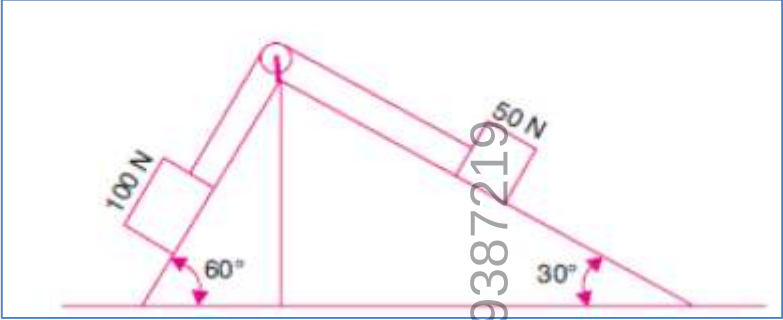
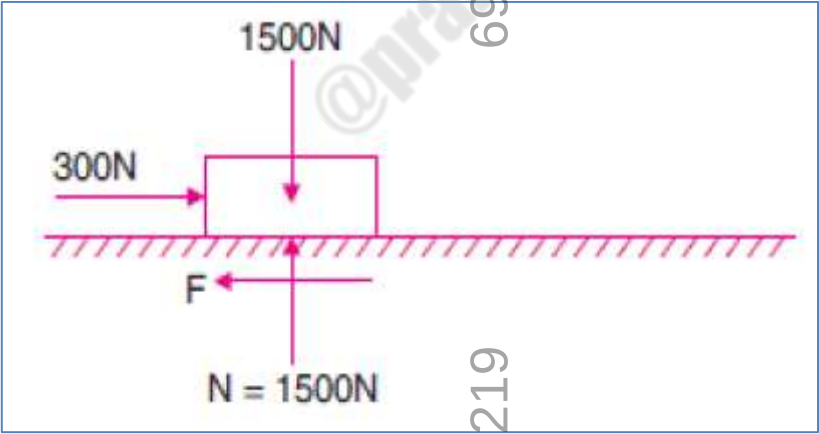
Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is NOT a type of support in mechanics?		1
	a) Fixed Support b) Roller Support c) Hinged Support d) Floating Support		
2	What is the centroid of a composite shape?		1
	a) The point where the shape can rotate freely. b) The center of gravity of the shape. c) The point of maximum stress. d) The point where all the forces are resolved.		
3	Lami's theorem is applicable to which type of force systems?		1
	a) Parallel forces b) Coplanar non-concurrent forces c) Coplanar concurrent forces d) None of the above		
4	Which method is NOT used to analyze a truss?		1
	a) Method of Joints b) Method of Sections c) Method of Nodes d) Graphical Method		
5	Which of the following is a motion with constant acceleration?		1
	a) Projectile motion b) Circular motion c) Uniform rectilinear motion d) Vibrational motion		
6	The instantaneous center of rotation of a rigid body is a point where:		1
	a) Acceleration is maximum. b) Velocity is zero. c) Force is maximum. d) Angular momentum is conserved.		
7	D'Alembert's principle is used to:		1
	a) Analyze static systems. b) Convert a dynamic problem into a static one. c) Study impulse and momentum. d) Analyze rolling resistance.		

8	What does the mass moment of inertia represent?					1
	a) Resistance to linear motion.	b) Resistance to rotational motion about an axis.	c) Center of gravity of a body.	d) The work done by a force.		
9	Which of the following is a scalar quantity?					1
	a) Work	b) Force	c) Momentum	d) Impulse		
10	The work-energy principle states that:					1
	a) Work is equal to force times acceleration.	b) The net work done is equal to the change in kinetic energy.	c) Potential energy is always conserved.	d) Energy can only be created, not destroyed.		
11	What is the unit of power in the SI system?					1
	a) Joule	b) Newton	c) Watt	d) Pascal		
12	In an elastic collision, which of the following quantities is conserved?					1
	a) Only momentum	b) Only kinetic energy	c) Both momentum and kinetic energy	d) Neither momentum nor kinetic energy		
Q.2	Solve the following.					12
A)	Explain system of forces and draw neat sketches for representing the system of forces.					6
B)	Determine the resultant of the three forces acting on a hook as shown in Fig. below.					6
						
Q.3	Solve the following.					12
A)	Draw Free body diagram of the reacting bodies (Ball, Ladder and Block weighing 600 N respectively)					6
						

B)	Find the forces in all the members of the truss shown in Fig. below .Tabulate the results. 	6
Q. 4	Solve Any Two of the following.	12
A)	A pilot flying his bomber at a height of 2000 m with a uniform horizontal velocity of 600 kmph wants to strike a target. At what distance from the target,he should release the bomb ? 	6
B)	Derive the relationship between angular motion and linear motion given by $a_n = \frac{v^2}{r} = r\omega^2$ Where Denoting radial acceleration by a_n , linear velocity by v, distance of the particle from the centre of rotation by r .	6
C)	The rotation of a fly wheel is governed by the equation $\omega = 3t^2 - 2t + 2$, where ω is in radian per second and t is in seconds. After one second from the start the angular displacement was 4 radians. Determine the angular displacement, angular velocity and angular acceleration of the fly wheel when $t = 3$ seconds.	6
Q.5	Solve Any Two of the following.	12
A)	Two ships move from a port at the same time. Ship A has velocity of 30 kmph and is moving in N 300 W while ship B is moving in south-west direction with a velocity of 40 kmph. Determine the relative velocity of A with respect to B and the distance between them after half an hour	6
B)	Determine the mass moment of inertia of a circular ring of uniform cross-section.	6
C)	Two rough planes inclined at 30° and 60° to horizontal are placed back to back shown in figure. The blocks of weights 50 N and 100 N	6

69387219	are placed on the faces and are connected by a string running parallel to planes and passing over a frictionless pulley. If the coefficient of friction between the planes and blocks is $\frac{1}{3}$, find the resulting acceleration and the tension in the string. 	69387219	
Q. 6	Solve Any Two of the following.		12
A)	State and prove work energy principle.		6
B)	Define following along with proper figure. a) Direct Impact b) Line of impact c) Oblique Impact d) Central Impact e) Eccentric Impact		6
C)	A 1500 N block is in contact with a level plane, the coefficient of friction between two contact surfaces being 0.1. If the block is acted upon by a horizontal force of 300 N, what time will elapse before the block reaches a velocity of 16 m/s starting from rest? If 300 N force is then removed, how much longer will the block continue to move? Solve the problem using the impulse momentum equation. 		6
	*** End ***		